



HYBRID ARCHITECTURE AND DEVELOPMENT FOR EXPERIMENTAL SYSTEMS (HADES) SERVICES

JAGUAR PERFORMANCE WORK STATEMENT (PWS)

**Space Systems Command (SSC)
Systems Delta 89 (SYD89)
Innovation and Prototyping (SD)
Prototype Ground Development and Sustainment
Branch (SDG)
Kirtland AFB, New Mexico**

Change Log

Version	Date	Summary of Changes
1.0	7 January 2026	Initial Draft

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1. PURPOSE

This Performance Work Statement (PWS) supplements the originating Hybrid Architecture and Development for Experimental Systems (HADES) Indefinite Delivery/Indefinite Quantity (IDIQ) PWS, specifically expanding on the Prototype Ground Development and Sustainment Branch's (Space Development Ground Support, or SDG) requirements for system sustainment and maintaining system stability. The HADES contract is a procurement for prototype and operational satellite ground systems, satellite operations centers (SOCs), and support services. Unless explicitly stated otherwise in this document, all aspects of the original HADES IDIQ PWS are still valid and applicable. As supplemental, the contractor shall perform satellite ground system and satellite operations center engineering, development, integration, and sustainment, and associated support services primarily at the RSC located at Kirtland, Air Force Base in New Mexico. Research, Development, Test, and Evaluation facility and may include occasional travel to contractor facilities for collaboration and integration testing.

2. BACKGROUND AND ASSUMPTIONS

The RSC requires secure, high-performance signal extension and switching capabilities to support critical missions. The organization's goal is to accelerate mission design and ground integration, launch and on-orbit operations, ground system tests, and rapid fielding to provide reliable, low-cost access to space in support of the warfighter. Ground support maintains infrastructure, systems, IT, cybersecurity and integration activities that are critical to day-to-day mission execution.

The Jaguar mission is currently managed out of the Jaguar Program Management Office but is subject to change in the future. The contractor under this task order is responsible for the sustainment of the Jaguar ground system and supporting infrastructure at the Satellite Operations Center (SOC) and Payload Operations Center (POC). Contractor direction, guidance, and oversight will be provided by the Government Task Order (TO) lead.

The Jaguar mission resides within the RSC and occupies Payload Test Center (PTC) 2, 3, & 4. The term "POC" refers to PTC-2, PTC-3 and PTC-4. The POC houses the Black Ground Configuration Controller (GCC), Red GCC, and GUTS and EC2K MMSOC Merged Architecture (GEMMA).

Personnel supporting Jaguar that require access to the POC require a Top Secret clearance with SCI eligibility. SCI clearance must be obtained within 120 days of hiring. Exceptions for support personnel (logistics, finance, scheduling, etc.) will be agreed upon and coordinated between the Government and the contractor on a case-by-case basis. The contractor bears all financial responsibility for obtaining clearances and SCI eligibility for their personnel.

The contractor will appoint a Task Order Manager to manage the work under this PWS to interface with the Government team. Travel (CONUS and OCONUS) is estimated to be <5 days per quarter.

3. OBJECTIVES

The contractor shall support the Jaguar mission with:

- Continued engineering, development, and sustainment of the SOC ground system
- Continued engineering, development, and sustainment of the POC ground system
- Continued development and support for program test and integration activities
- Continued sustainment support for external customers
- Planning, procurement, development, installation, testing, and operational readiness for the Jaguar ground systems
- The development and sustainment activities of the OCONUS Ground Terminals (GTs) to include but not limited to:
 - Remote troubleshooting
 - Remote patching **and** software upgrades
 - Supporting infrastructure maintenance & upgrades

The contractor shall work with SSC/SYD89/SD Government, Jaguar Program Management Office, Prototype Operations I (POPS-I), the Space Vehicle contractor (SV Ktr) personnel, and all other external stakeholders to ensure POC and SOC ground system readiness satisfies requirements outlined in this document.

The PWS requirements from the originating HADES PWS which are applicable to this effort are identified via bullet lists below. In general, the following paragraphs apply only to the extent the work is directly applicable to the accomplishment of this effort. Further clarified specifications are identified for specific requirements where appropriate.

4. BASE EFFORT AND OPTIONS

The base effort and options include engineering, development, and sustainment of the Payload Operations Center (POC)/Satellite Operations Center (SOC) ground systems to include:

- GCC
- GEMMA
- Architecture supporting spacecraft bus and payload TT&C (MUS software)
 - Limited to HW infrastructure, front end processors, crypto devices, network equipment, high speed guard (HSG), and any other peripheral equipment necessary to conduct satellite bus Telemetry, Tracking, and Commanding (TT&C) operations
 - May require government delivery of TT&C SW and/or MUS
- Compunetix phone system
- Limited support to the Alternate Ground Terminal (AGT) and OCONUS GTs (as defined herein, paragraph 8.1)

4.1 The contractor shall support the following on all assets accepted by the Government for sustainment:

- Continual information system security engineer (ISSE) work
- Single point of failure analysis
- Automation efforts
- Continuous improvement efforts

4.2 The contractor shall provide hardware support for Compunetix system integration

4.3 The contractor shall support Compunetix installation instructions for any required functionality, the contractor is not expected to provide infrastructure support/materials or personnel to support installs at locations outside the POC.

4.4 The contractor shall provide Compunetix support, to include depot support, during normal business hours at the POC (0800-1600 MST)

4.5 The contractor shall proactively perform procurement, installation, integration, test, and deployment of new information assurance (IA) approved hardware, software, peripherals, and network equipment to eliminate any single points of failure or items identified as nearing End of Life (EOL) or End of Support (EOS) within SOC and POC boundary.

4.6 The contractor shall procure all Jaguar hardware and software in accordance with the Government-approved Bill of Materials (BOM), to be approved by the Government Mission Manager.

4.7 The contractor shall provide on-location and over-the-phone support for external customers as needed including, but not limited to support for: Compunetix, Integrated System Testing (IST), GCC and procurement at the Payload Hosted In GCC (PHIG) and Government Furnished Equipment Provider (GFEP), and any network connection troubleshooting that falls within the POC and/or GEMMA boundaries.

4.8 The contractor shall manage the storage and accountability of hardware components for the Jaguar mission on Kirtland Air Force Base (RSC and MRF). The contractor shall assist appropriate government representative(s) with shipping these component to other CONUS or OCONUS locations as directed.

4.9 The contractor shall manage the accountability of Compunetix equipment under their purview. Equipment shipped to geographically separated locations shall be documented via a signed DD Form 1149.

4.10 The contractor shall sustain PTC 2, 3, & 4 hardware and infrastructure and perform technical refreshes as needed. The contractor shall procure the appropriate HW, SW, and peripherals as necessary per the security standard to meet all ATO and Sensitive Compartmented Information Facility (SCIF) requirements. The contractor shall provide escorts and coordinate with the government to ensure escort requirements do not impact scheduled work.

4.11 The contractor shall provide on-site support beginning at contract award.

- System Administrators are required to be on-site 16 hours a day, 5 days a week (16x5)

- All other functions are required to be on-site during core business hours, 5 days a week (8x5)

4.12 The contractor shall provide 24x7 on-call support with a 2-hour response time.

4.13 The contractor shall participate, at a minimum, in all meeting listed in Table 4.1

Table 4.1, Contractor Supported Meetings

MEETING TITLE	DURATION	FREQUENCY
Contractor/Government Weekly Tag up	1 hour	Weekly
Ops and Sustainment Weekly Tag up	1 hour	Weekly
Integrated Product Team Tag-up	1.5 hours	Weekly
Program Office Update	1.5 hours	Weekly
External Customer Update	1.5 hours	Weekly

5. SYSTEM AVAILABILITY REQUIREMENT

5.1 The POC ground systems availability shall be at least 99% in any sliding 30-day window. The SOC ground system availability shall be at least 95% in any sliding 30-day window. Down time required for Jaguar specific PMIs, installs, and other Government approved events will not count against availability requirements.

5.2 System Availability is further defined as follow:

- Unscheduled, or any other downtime that did not have pre-coordinated government approval will count against the 99% POC requirement and 95% SOC requirement.
- There are approximately six SCN contacts per vehicle, per day which should count towards the SOC system availability requirement. All required/scheduled payload operations will count against the POC system availability requirement.

5.3 Response time to any system failure by the responding contractor contingent shall be less than 2 hours after notification.

5.4 The contractor shall provide Weekly & Monthly system availability reports. These reports shall include metrics calculated with and without PMI downtime.

6. SCOPE

The following requirements are applicable to this effort:

▪ 1.2 Scope	
▪ 1.2.1	▪ 1.2.2
▪ 1.2.3	▪ 1.2.4

▪ 1.2.5	▪ 1.2.6
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7. ACCESS TO COMPUTER SYSTEMS

▪ 1.3.1	
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8. BASIC SERVICES

▪ 2.0 Basic Services	
▪ 2.1 Contract Transition	
▪ 2.1.1	▪ 2.1.2
▪ 2.2 Support Services	
▪ 2.2.1 Program Management	
▪ 2.2.1.1	▪ 2.2.1.2
▪ 2.2.1.3	▪ 2.2.1.4
▪ 2.2.1.5	▪ 2.2.1.6
▪ 2.2.1.7	▪ 2.2.1.8
▪ 2.2.1.9	▪ 2.2.1.10
▪ 2.2.1.11	▪ 2.2.1.12
▪ 2.2.2 Cost Management	
▪ 2.2.2.1	▪ 2.2.2.2
▪ 2.2.2.3	▪ 2.2.2.4
▪ 2.2.2.5	
▪ 2.2.3 Information Management	
▪ 2.2.3.1	▪ 2.2.3.2
▪ 2.2.3.3	
▪ 2.2.4 Security and Data Handling	
▪ 2.2.4.1	▪ 2.2.4.2
▪ 2.2.4.3	▪ 2.2.4.4
▪ 2.2.4.5	▪ 2.2.4.6
▪ 2.2.4.7	▪ 2.2.4.8

▪ 2.2.4.9	▪ 2.2.4.10
▪ 2.2.4.11	▪ 2.2.4.12
▪ 2.2.4.13	▪ 2.2.4.14
▪ 2.2.4.15	
▪ 2.2.5 Quality Control	
▪ 2.2.5.1.	▪ 2.2.5.2
▪ 2.2.5.3	▪ 2.2.5.4
▪ 2.2.5.5	▪ 2.2.5.6
▪ 2.2.6 Risk Management	
▪ 2.2.6.1	▪ 2.2.6.1
▪ 2.2.6.3	
▪ 2.2.7 Training and Certification	
▪ 2.2.7.1.	▪ 2.2.7.2
▪ 2.2.7.3	▪ 2.2.7.4
▪ 2.2.7.5	▪ 2.2.7.6
▪ 2.2.7.7	▪ 2.2.7.8
▪ 2.2.7.9	▪ 2.2.7.10
▪ 2.2.7.11	
▪ 2.2.8 Configuration Management	
▪ 2.2.8.1	▪ 2.2.8.2
▪ 2.2.8.3	▪ 2.2.8.4
▪ 2.2.9 Procurement, Logistics, and Lifecycle Management	
▪ 2.2.9.1.	▪ 2.2.9.2
▪ 2.2.9.3	▪ 2.2.9.4
▪ 2.2.9.5	▪ 2.2.9.6
▪ 2.2.9.7	▪ 2.2.9.8
▪ 2.2.9.9	▪ 2.2.9.10
▪ 2.2.9.11	▪ 2.2.9.12

▪ 2.2.9.13	▪ 2.2.9.14
▪ 2.2.9.15	▪ 2.2.9.16
▪ 2.2.9.17	▪ 2.2.9.18
▪ 2.2.9.19	▪ 2.2.9.20
▪ 2.2.9.21	▪ 2.2.9.22
▪ 2.2.9.23	▪ 2.2.9.24
▪ 2.2.9.25	▪ 2.2.9.26
▪ 2.2.9.27	▪ 2.2.9.28
▪ 2.2.9.29	▪ 2.2.9.30
▪ 2.2.10 Concept Development	
▪ 2.2.10.1	▪ 2.2.10.2
▪ 2.2.10.3	▪ 2.2.10.4
▪ 2.2.10.5	▪

9. GROUND SYSTEM SERVICES

▪ 2.3 Ground System Services	
▪ 2.3.1 Ground System Engineering and Design	
▪ 2.3.1.1	▪ 2.3.1.2
▪ 2.3.1.3	▪ 2.3.1.4
▪ 2.3.1.5	▪ 2.3.1.6
▪ 2.3.1.7	▪ 2.3.1.8
▪ 2.3.1.9	▪ 2.3.1.10
▪ 2.3.2 Ground System Development	
▪ 2.3.2.1	▪ 2.3.2.2
▪ 2.3.2.3	▪ 2.3.2.4
▪ 2.3.2.5	▪ 2.3.2.6
▪ 2.3.2.7	▪ 2.3.2.8

▪ 2.3.3 Ground System Integration	
▪ 2.3.3.1	▪ 2.3.3.2
▪ 2.3.3.3	▪ 2.3.3.4
▪ 2.3.3.5	▪ 2.3.3.6
▪ 2.3.3.7	▪ 2.3.3.8
▪ 2.3.3.9	▪ 2.3.3.10
▪ 2.3.3.11	▪ 2.3.3.12
▪ 2.3.3.13	▪ 2.3.3.14
▪ 2.3.4 Ground System Sustainment	
▪ 2.3.4.1	▪ 2.3.4.2
▪ 2.3.4.3	▪ 2.3.4.4
▪ 2.3.4.5	▪ 2.3.4.6
▪ 2.3.4.7	▪ 2.3.4.8
▪ 2.3.4.9	
▪ 2.3.5 Ground System Decommission, Disposal, and/or Transfer	
▪ 2.3.5.1	▪ 2.3.5.2
▪ 2.3.5.3	▪ 2.3.5.4
▪ 2.3.5.5	▪ 2.3.5.6
▪ 2.3.5.7	▪ 2.3.5.8
▪ 2.3.5.9	

8.1 GTs and AGTs

This section covers Ground Terminal antenna support for remote system administrator and network engineer support that may be required.

8.1.1 The contractor shall provide remote support from Kirtland AFB to assist troubleshooting and maintenance of all OCONUS GTs and AGTs.

OCONUS GT and AGT support will include:

- Provide government approved logs & screenshots.
- Implement government approved configuration changes.

- Install government approved software & patch updates.
- Provide government approved account management and network maintenance.
- Provide necessary inputs/products to the government for configuration management or procurement consideration.
- Provide remote support in prevention and corrective maintenance.
- Provide in-person support on a case-by-case, special circumstance basis.

This level of effort will be scrutinized by the Government TO lead to ensure the level of effort is consistent with the objective of this PWS.

10. OPERATIONAL SUPPORT

10.1 Provide Hardware, Software, System Administration, Network Engineer, and COMSEC support to keep the Jaguar systems operational. Hardware and software purchases shall be approved by the Government prior to the purchase. All contractor functional groups defined above may be required to support during off-hours, on a case-by-case basis.

10.2 Maintain and sustain the SOC and POC ground systems/subsystems required to support the Jaguar mission. This includes sustainment support for Jaguar mission unique aspects of the hardware and software.

10.3 Develop procedures that will aid in recovering from ground system maintenance, anomalies, and pre-/post- high-tempo operations.

10.4 Ensure all Jaguar SOC and POC ground system infrastructure is mission capable to provide support to operational crews preparing for and executing development, test, verification, validation, and operational events.

10.5 Coordinate with operations teams to ensure all required pre and post activities for development, test, verification, validation, and high-tempo operational events are executed when required.

10.6 Verify and ensure that all mission databases are in an optimal state, redundant, and in-sync for data redundancy. Update/create procedures for transition between the domains minimizing down time of the system. This includes procedures for any changes to the Cross Domain Solution (CDS) required for successful domain switching.

10.7 Provide over the phone support during core business hours for the Compunetix phone system, Long Line Video Distribution (LLVD), File share, and VPOAT to all applicable internal and external users on the Jaguar program. If over the phone support cannot resolve the issue the contractor shall have the equipment shipped back for inspection or repaired from on-hand spares. The contractor shall consult with Compunetix or software reach back support in all efforts their team cannot resolve.

10.8 Maintain a complete set of spares and repair inventory and audit of that equipment for the Compunetix system.

11. SUPPORTING INFRASTRUCTURE

▪ 2.4.1 Power Distribution and Resiliency Support	
▪ 2.4.1.1	2.4.1.2
▪ 2.4.1.3	2.4.1.4
▪ 2.4.1.5	2.4.1.6
▪ 2.4.2 Heating, Ventilation and Air Conditioning Support	
▪ 2.4.2.1	2.4.2.2
▪ 2.4.2.3	2.4.2.4
▪ 2.4.2.5	2.4.2.6
▪ 2.4.3 Facility and Furniture Maintenance	
▪ 2.4.3.1	2.4.3.2
▪ 2.4.3.3	2.4.3.4
▪ 2.4.3.5	2.4.3.6
▪ 2.4.4 Network Infrastructure Support	
▪ 2.4.4.1	2.4.4.3
▪ 2.4.4.5	▪ 2.4.4.6
▪ 2.4.5 Space Optimization Layout	
▪ 2.4.5.1	▪ 2.4.5.2
▪ 2.4.5.3	▪ 2.4.5.4
▪ 2.4.5.5	▪ 2.4.5.6

12. SOFTWARE SUPPORT SERVICES

▪ 2.5.1 Software Setup	
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▪ 2.5.1.1	▪ 2.5.1.2
▪ 2.5.1.3	▪ 2.5.1.4
▪ 2.5.2 Software Development	
▪ 2.5.2.1	▪ 2.5.2.2
▪ 2.5.2.3	▪ 2.5.2.4
▪ 2.5.2.5	▪ 2.5.2.6
▪ 2.5.2.7	▪ 2.5.2.8
▪ 2.5.3 Software Configuration	
▪ 2.5.3.1	▪ 2.5.3.2
▪ 2.5.3.3	

13. Other Activities

▪ 2.6 Other Activities	
▪ 2.6.1 COMSEC	
▪ 2.6.1.1	▪ 2.6.1.2
▪ 2.6.1.3	▪ 2.6.1.4
▪ 2.6.1.5	▪ 2.6.1.6
▪ 2.6.1.7	▪ 2.6.1.8
▪ 2.6.1.9	▪ 2.6.1.10
▪ 2.6.1.11	▪ 2.6.1.12
▪ 2.6.1.13	
▪ 2.6.2 Cybersecurity	
▪ 2.6.2.1	▪ 2.6.2.2
▪ 2.6.2.3	▪ 2.6.2.4
▪ 2.6.2.5	▪ 2.6.2.6
▪ 2.6.2.7	▪ 2.6.2.8
▪ 2.6.2.9	▪
▪ 2.6.3 Defensive Cybersecurity Operations (DCO)	▪

▪ 2.6.3.1	▪ 2.6.3.2
▪ 2.6.3.3	▪ 2.6.3.4

13.1 Upon government request, the contractor shall provide special studies on current capabilities and with no less than three courses of action (COA) for system and architectural improvements and a recommended path forward for the system. At a minimum, the report will articulate the recommended path/COA forward, milestone schedule, and a Rough Order of Magnitude (ROM) cost for each COA.

13.2 The contractor shall package/ship hardware as needed in support of sustainment activities of the SOC, POC, Compunetix system and GT systems. The contractor will not ship to OCONUS locations. The contractor will transfer necessary OCONUS shipments to a government directed party. This includes sending equipment back to the vendor for return merchandizing authorizations (RMA), as well as shipping/receiving spare depot parts for OCONUS operations support.